

Asset Pricing in the Standard Nuclear oliGARCHy

From Systemic Risk to Risk Premia

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Abstract

This paper proposes an asset-pricing layer for the Standard Nuclear oliGARCHy (SNoG). The central objective is to connect a structural representation of systemic nuclear risk to equilibrium asset pricing, risk premia, volatility, portfolio choice, and financial-market feedback. The framework treats systemic risk as a dynamic state rather than as a static exogenous variable. Let the SNoG state be represented by a vector $\mathbf{X}_t$, with financial prices, geopolitical conditions, institutional variables, and market expectations jointly determining its evolution. A stochastic discount factor is then allowed to depend on the SNoG state, generating state-dependent risk premia.

The framework combines modern asset pricing, stochastic volatility, multivariate time-series econometrics, network theory, control theory, game theory, international political economy, welfare economics, decision theory, behavioural economics, computational statistics, machine learning, and systemic-risk analysis. The paper develops testable propositions concerning the relation between systemic nuclear risk and expected returns, volatility, correlations, liquidity, sovereign spreads, currencies, commodities, and cross-market contagion.

The proposed architecture does not assume that the SNoG necessarily possesses incremental explanatory power. Instead, it establishes a falsifiable empirical programme: estimate conventional financial models, augment them with SNoG state variables, and test whether the additional structure produces statistically significant, economically meaningful, and out-of-sample improvements.

The paper ends with “The End”

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1 Introduction

The principal contribution proposed in this paper is an explicit financial valuation layer for the Standard Nuclear oliGARCHy (SNoG). The underlying intuition is straightforward. Systemic geopolitical risk can affect economic agents, but financial markets also transform expectations about systemic risk into prices, quantities, volatility, liquidity, and capital allocation.

Consequently, the relationship should not necessarily be represented as a one-directional causal chain,

$$\text{Geopolitical Risk} \longrightarrow \text{Financial Markets.} \quad (1)$$

A more general representation is a coupled dynamical system,

$$\mathbf{X}_{t+1} = F(\mathbf{X}_t, \mathbf{P}_t, \mathbf{G}_t, \mathbf{M}_t, \varepsilon_{t+1}), \quad (2)$$

where $\mathbf{X}_t$ denotes the systemic state, $\mathbf{P}_t$ financial prices, $\mathbf{G}_t$ geopolitical variables, and $\mathbf{M}_t$ macroeconomic variables.

Financial prices themselves depend upon expectations concerning the future state,

$$\mathbf{P}_t = G(\mathbf{X}_t, \mathbb{E}_t[\mathbf{D}_{t+1}], \mathbb{E}_t[\mathbf{D}_{t+2}], \dots). \quad (3)$$

The resulting system is reflexive: the state affects prices while prices can alter the state through financing conditions, wealth effects, political incentives, institutional stress, and strategic responses.

The purpose of this paper is therefore not to declare that nuclear systemic risk has a particular financial price. Rather, it asks a testable question:

Does information contained in an SNoG state representation explain financial risk and risk premia beyond information already contained in conventional asset-pricing and macro-financial models?

2 Conceptual Architecture

2.1 The SNoG State

Let the systemic state be

$$\mathbf{X}_t = [x_{1,t} \quad x_{2,t} \quad \cdots \quad x_{K,t}]'. \quad (4)$$

The components may represent quantities such as:

1. strategic tension;
2. alliance configuration;
3. military escalation indicators;
4. nuclear posture indicators;
5. diplomatic stress;
6. sanctions;
7. energy dependence;
8. sovereign financial stress;

9. market volatility;
10. liquidity conditions;
11. expectations and uncertainty.

The state need not be directly observable. It may instead be a latent variable estimated from observable indicators.

A general state-space representation is

$$\mathbf{X}_{t+1} = F(\mathbf{X}_t, \mathbf{Z}_t, \varepsilon_{t+1}), \quad (5)$$

$$\mathbf{Y}_t = H(\mathbf{X}_t, \eta_t), \quad (6)$$

where $\mathbf{Y}_t$ denotes observed indicators and $\mathbf{Z}_t$ denotes observable exogenous or predetermined variables.

2.2 Financial Prices as State Variables

Let $\mathbf{P}_t$ denote a vector containing prices of:

- equities;
- sovereign bonds;
- corporate bonds;
- currencies;
- commodities;
- derivatives;
- credit instruments.

The system may be represented schematically as follows.

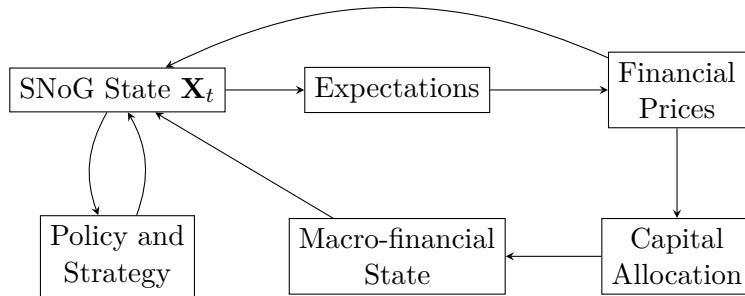


Figure 1: System diagram.

The diagram emphasizes that the proposed system is not merely a risk-index construction. It is a feedback architecture.

3 Asset Pricing Foundation

3.1 Stochastic Discount Factor

Let m_{t+1} denote a stochastic discount factor. For a traded asset with gross return $R_{i,t+1}$,

$$1 = \mathbb{E}_t[m_{t+1}R_{i,t+1}]. \quad (7)$$

For the risk-free asset,

$$1 = \mathbb{E}_t[m_{t+1}R_{f,t}]. \quad (8)$$

Therefore,

$$\mathbb{E}_t[R_{i,t+1}] - R_{f,t} = -\text{Cov}_t(m_{t+1}, R_{i,t+1}). \quad (9)$$

This equation provides the fundamental bridge between the SNoG and asset pricing.

3.2 State-Dependent Discount Factor

Suppose

$$m_{t+1} = m(\mathbf{X}_t, \mathbf{X}_{t+1}, \mathbf{M}_{t+1}). \quad (10)$$

Then

$$\mathbb{E}_t[R_{i,t+1}] - R_{f,t} = -\text{Cov}_t(m(\mathbf{X}_t, \mathbf{X}_{t+1}, \mathbf{M}_{t+1}), R_{i,t+1}). \quad (11)$$

Thus the risk premium becomes conditional on the systemic state.

Definition 1. A SNoG risk premium is the component of an asset's conditional expected excess return associated with its covariance with the SNoG-dependent stochastic discount factor.

3.3 Linear Factor Representation

Suppose the stochastic discount factor admits the approximation

$$m_{t+1} = a_t - \boldsymbol{\lambda}'_t \mathbf{f}_{t+1}, \quad (12)$$

where $\mathbf{f}_{t+1}$ is a vector of risk factors.

Let the factors include an SNoG factor S_{t+1} . Then

$$\mathbf{f}_{t+1} = \begin{bmatrix} R_{M,t+1} - R_{f,t} \\ S_{t+1} \\ V_{t+1} \\ L_{t+1} \end{bmatrix}. \quad (13)$$

Expected excess returns can then be written as

$$\mathbb{E}_t[R_{i,t+1} - R_{f,t}] = \beta'_{i,t} \boldsymbol{\lambda}_t. \quad (14)$$

In particular,

$$\mathbb{E}_t[R_{i,t+1} - R_{f,t}] = \beta_{M,i,t} \lambda_{M,t} + \beta_{S,i,t} \lambda_{S,t} + \beta_{V,i,t} \lambda_{V,t} + \beta_{L,i,t} \lambda_{L,t}. \quad (15)$$

The coefficient $\lambda_{S,t}$ is the conditional price of SNoG systemic risk.

4 A Structural Definition of SNoG Exposure

Define an asset's SNoG exposure as

$$\beta_{S,i,t} = \frac{\text{Cov}_t(R_{i,t+1}, S_{t+1})}{\text{Var}_t(S_{t+1})}. \quad (16)$$

This definition does not imply that positive exposure is desirable or undesirable. It merely defines statistical sensitivity.

The economic interpretation depends upon the payoff structure.

An asset whose return falls when systemic risk increases has negative covariance with a risk state that represents adverse conditions. An asset whose payoff increases under the same conditions may have positive covariance.

5 Systemic Risk and Volatility

A multivariate conditional covariance model may be written as

$$\mathbf{r}_t = \boldsymbol{\mu}_t + \mathbf{H}_t^{1/2} \mathbf{z}_t, \quad (17)$$

where $\mathbf{r}_t$ is the return vector and $\mathbf{H}_t$ is the conditional covariance matrix.

A SNoG-augmented volatility specification is

$$h_{i,t} = \omega_i + \alpha_i \epsilon_{i,t-1}^2 + \beta_i h_{i,t-1} + \gamma_i S_{t-1}, \quad (18)$$

$$S_t = \Phi(S_{t-1}, \mathbf{G}_{t-1}, \mathbf{M}_{t-1}, \mathbf{P}_{t-1}) + u_t. \quad (19)$$

The parameter γ_i measures the incremental conditional variance response to the systemic state.

6 OliGARCH Interpretation

The SNoG may be connected to generalized autoregressive conditional heteroskedasticity through the following generic structure:

$$\mathbf{H}_t = \boldsymbol{\Omega} + \sum_{j=1}^p \mathbf{A}_j \epsilon_{t-j} \epsilon'_{t-j} \mathbf{A}'_j + \sum_{j=1}^q \mathbf{B}_j \mathbf{H}_{t-j} \mathbf{B}'_j + \boldsymbol{\Gamma} \mathbf{X}_{t-1} \mathbf{X}'_{t-1} \boldsymbol{\Gamma}'. \quad (20)$$

The final term allows systemic-state variables to affect conditional covariance.

The empirical question is whether $\boldsymbol{\Gamma}$ contains information not already captured by the conventional GARCH dynamics.

7 Network Representation

Let the SNoG be represented as a directed weighted graph

$$\mathcal{G}_t = (\mathcal{V}, \mathcal{E}, \mathbf{W}_t). \quad (21)$$

Nodes may represent states, institutions, markets, strategic actors, or financial systems. Edge weights represent measurable dependence.

Let

$$W_{ij,t} = f_{ij}(\text{trade, finance, alliances, energy, strategic exposure}). \quad (22)$$

A financial contagion matrix may then be defined as

$$\mathbf{C}_t = \mathbf{D}_t^{-1/2} \mathbf{W}_t \mathbf{D}_t^{-1/2}. \quad (23)$$

The eigenstructure of $\mathbf{C}_t$ provides one possible measure of network concentration and systemic interconnectedness.

8 Physics and Dynamical Systems

The system can be interpreted as a nonlinear dynamical system,

$$\mathbf{X}_{t+1} = F(\mathbf{X}_t). \quad (24)$$

A local approximation around a fixed point $\mathbf{X}^*$ gives

$$\Delta \mathbf{X}_{t+1} \approx \mathbf{J}_F(\mathbf{X}^*) \Delta \mathbf{X}_t, \quad (25)$$

where $\mathbf{J}_F$ is the Jacobian matrix.

If the spectral radius satisfies

$$\rho(\mathbf{J}_F(\mathbf{X}^*)) < 1, \quad (26)$$

the local linearized system is dynamically stable.

If instead

$$\rho(\mathbf{J}_F(\mathbf{X}^*)) > 1, \quad (27)$$

small perturbations can amplify locally.

This provides a mathematical language for distinguishing perturbations from self-reinforcing feedback.

9 Chemistry and Reaction-Network Analogy

A systemic crisis can also be represented metaphorically as a reaction network. Let states be nodes and transitions be reactions,



Financial stress can alter transition rates,

$$k_{ij} = k_{ij}(\mathbf{X}_t). \quad (29)$$

The analogy is useful only insofar as it provides a formal framework for nonlinear transition rates. It should not be interpreted as claiming that geopolitical systems literally obey chemical kinetics.

10 Biological and Epidemiological Analogy

Financial contagion has structural similarities to epidemic propagation.

Let I_t denote the number or intensity of distressed financial nodes. A simplified model is

$$\Delta I_t = \beta_t S_t I_t - \gamma_t I_t, \quad (30)$$

$$\beta_t = \beta_0 + \beta_1 S_t^{\text{SNoG}}. \quad (31)$$

Here β_t represents the transmission coefficient and γ_t the recovery rate.

The analogy suggests examining whether systemic risk changes the effective reproduction number,

$$\mathcal{R}_t = \frac{\beta_t}{\gamma_t}. \quad (32)$$

A value exceeding unity would represent amplification in this stylized model.

11 Psychology, Decision Theory, and Behavioural Finance

Financial agents do not necessarily form expectations through a single rational mechanism.

Let subjective expectations be

$$\mathbb{E}_t^i[X_{t+1}] = \mathbb{E}_t[X_{t+1}] + b_{i,t}, \quad (33)$$

where $b_{i,t}$ represents agent-specific belief distortion.

A behavioural extension may specify

$$b_{i,t} = \psi_i(S_t, \mathcal{H}_{i,t}, \mathcal{L}_{i,t}), \quad (34)$$

where $\mathcal{H}_{i,t}$ represents historical experience and $\mathcal{L}_{i,t}$ information or learning.

Possible mechanisms include:

- ambiguity aversion;
- loss aversion;
- recency effects;
- herding;
- attention constraints;
- confirmation effects;
- probability distortion.

These mechanisms can affect the mapping from objective systemic risk to observed market prices.

12 Psychiatric Concepts and Scientific Boundaries

Concepts originating in psychiatry should not be used to diagnose market participants or political actors from financial data.

Instead, formally defined constructs such as uncertainty, risk perception, stress, attention, and behavioural response can be modelled without assigning clinical diagnoses.

For example,

$$A_t = a_0 + a_1 S_t + a_2 V_t + a_3 L_t + \nu_t \quad (35)$$

may represent an attention index estimated from observable behaviour.

The distinction between a measurable behavioural variable and a clinical diagnosis is essential for scientific validity.

13 Philosophical Foundations

The framework contains an epistemological distinction between:

1. an underlying systemic state;
2. observations generated by that state;
3. beliefs concerning that state;
4. decisions based upon those beliefs.

Formally,

$$\text{State} \longrightarrow \text{Observation} \longrightarrow \text{Belief} \longrightarrow \text{Decision}. \quad (36)$$

Financial prices are therefore neither necessarily identical to the underlying state nor independent observations of it.

This distinction is particularly important when the state itself is partially latent.

14 Political Science and Geopolitics

The geopolitical state can be represented by

$$\mathbf{G}_t = g(\text{military posture, alliances, diplomacy, sanctions, trade, strategic uncertainty}). \quad (37)$$

A strategic interaction can be expressed as a repeated game in which agents choose actions

$$a_{i,t} \in \mathcal{A}_i. \quad (38)$$

Their payoff is

$$U_i = U_i(a_{1,t}, \dots, a_{N,t}, \mathbf{X}_t). \quad (39)$$

The financial market responds not only to realized actions but also to expectations of future actions,

$$P_t = P(\mathbb{E}_t[a_{i,t+1}], \mathbb{E}_t[a_{i,t+2}], \dots). \quad (40)$$

15 Diplomacy and International Relations

Diplomatic information can be treated as an observable signal.

Let

$$D_t = d(\text{statements, negotiations, agreements, crises, communication intensity}). \quad (41)$$

The empirical task is to determine whether such information improves forecasting after controlling for financial and macroeconomic variables.

A neutral specification is

$$R_{i,t+1} = \alpha_i + \beta_i S_t + \delta_i D_t + \boldsymbol{\theta}'_i \mathbf{Z}_t + \epsilon_{i,t+1}. \quad (42)$$

16 Welfare Economics

The financial consequences of systemic risk can affect social welfare.

Let social welfare be

$$W_t = W(C_t, Y_t, L_t, R_t), \quad (43)$$

where consumption, output, liquidity, and systemic risk enter the welfare function.

A representative loss function may be

$$\mathcal{L} = \omega_Y(Y - Y^*)^2 + \omega_\pi(\pi - \pi^*)^2 + \omega_R(R - R^*)^2. \quad (44)$$

The weights ω_Y , ω_π , and ω_R are normative parameters rather than empirical constants.

Consequently, empirical identification of systemic risk does not by itself determine the appropriate policy response.

17 Warfare Economics

A stylized resource-allocation problem can be written as

$$Y_t = C_t + I_t + D_t + G_t, \quad (45)$$

where D_t represents defence expenditure.

An increase in strategic risk may alter the allocation constraint,

$$D_{t+1} = D_t + \phi(S_t). \quad (46)$$

The financial market consequences may include changes in:

- sovereign borrowing;
- taxation expectations;
- inflation expectations;
- commodity prices;
- defence-industry valuations;
- exchange rates;
- sovereign credit risk.

These are empirical hypotheses, not identities.

18 A Formal Feedback Model

The complete architecture can be represented as

$$\mathbf{X}_{t+1} = F(\mathbf{X}_t, \mathbf{P}_t, \mathbf{G}_t, \mathbf{M}_t, \varepsilon_{t+1}), \quad (47)$$

$$\mathbf{P}_t = G(\mathbf{X}_t, \mathbb{E}_t[\mathbf{D}_{t+1}], \mathbb{E}_t[\mathbf{D}_{t+2}], \dots), \quad (48)$$

$$\mathbf{H}_t = H(\mathbf{X}_t, \mathbf{r}_{t-1}, \mathbf{H}_{t-1}). \quad (49)$$

This creates three interconnected layers:

1. structural systemic dynamics;
2. financial valuation;
3. conditional risk dynamics.

19 Propositions

Proposition 1 (State-dependent pricing). *If the stochastic discount factor depends on the SNoG state and an asset's return covaries with that state, then its conditional expected excess return generally differs from the corresponding return under a model omitting that state, unless the additional covariance contribution is exactly zero or is already spanned by existing factors.*

Proof. From the stochastic discount-factor identity,

$$\mathbb{E}_t[R_{i,t+1}] - R_{f,t} = -\text{Cov}_t(m_{t+1}, R_{i,t+1}). \quad (50)$$

Suppose

$$m_{t+1} = m_{t+1}^0 + m_{t+1}^S. \quad (51)$$

Then

$$-\text{Cov}_t(m_{t+1}, R_{i,t+1}) = -\text{Cov}_t(m_{t+1}^0, R_{i,t+1}) - \text{Cov}_t(m_{t+1}^S, R_{i,t+1}). \quad (52)$$

The second term is nonzero whenever the SNoG-dependent component of the discount factor has nonzero covariance with the asset return. Therefore the conditional expected excess return contains an additional pricing component unless that component is zero or redundant with existing factors. $\square$

Proposition 2 (Incremental empirical value). *An SNoG variable provides incremental empirical value only if its inclusion improves a prespecified statistical or economic criterion relative to a correctly specified benchmark model.*

Proof. Let M_0 be a benchmark model and M_1 an augmented SNoG model. Define a loss function L .

The out-of-sample improvement is

$$\Delta L = L(M_0) - L(M_1). \quad (53)$$

If $\Delta L > 0$ under an appropriate evaluation criterion, the augmented model provides predictive improvement according to that criterion. If $\Delta L \leq 0$, the SNoG variable does not provide predictive improvement according to that criterion.

Thus the claim is inherently empirical. $\square$

Proposition 3 (Feedback amplification). *Consider the local system*

$$\Delta \mathbf{X}_{t+1} = \mathbf{J} \Delta \mathbf{X}_t. \quad (54)$$

If the spectral radius of $\mathbf{J}$ exceeds unity, there exists at least one local direction in which perturbations can grow under the linearized dynamics.

Proof. The asymptotic behaviour of the linear system is determined by the eigenvalues of $\mathbf{J}$. If an eigenvalue λ satisfies

$$|\lambda| > 1, \quad (55)$$

the corresponding eigenmode grows geometrically as

$$\lambda^t. \quad (56)$$

Hence the system contains an amplifying local direction. $\square$

20 Econometric Identification

The principal regression is

$$R_{i,t+1} - R_{f,t} = \alpha_i + \beta'_i \mathbf{F}_t + \gamma_i S_t + \epsilon_{i,t+1}. \quad (57)$$

The null hypothesis is

$$H_0 : \gamma_i = 0. \quad (58)$$

The alternative is

$$H_1 : \gamma_i \neq 0. \quad (59)$$

However, statistical significance alone is insufficient.

A complete evaluation should consider:

1. coefficient significance;
2. economic magnitude;
3. parameter stability;
4. robustness to alternative specifications;
5. out-of-sample forecasting;
6. portfolio implications;
7. model uncertainty.

21 Panel Specification

For countries or assets indexed by i ,

$$R_{i,t+1} = \alpha_i + \delta_t + \beta S_{i,t} + \boldsymbol{\theta}' \mathbf{Z}_{i,t} + u_{i,t+1}. \quad (60)$$

Here α_i represents entity fixed effects and δ_t time effects.

A dynamic specification is

$$R_{i,t+1} = \alpha_i + \delta_t + \rho R_{i,t} + \beta S_{i,t} + \boldsymbol{\theta}' \mathbf{Z}_{i,t} + u_{i,t+1}. \quad (61)$$

Potential identification problems include:

- omitted variables;
- reverse causality;
- measurement error;
- simultaneous determination;

- structural breaks;
- nonstationarity;
- cross-sectional dependence.

22 Causal Inference

The structural causal question can be represented by a directed acyclic graph in an appropriately defined empirical setting.

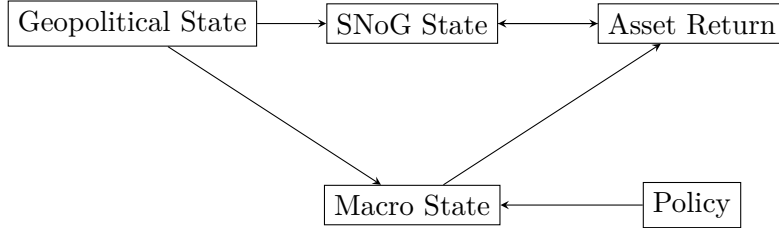


Figure 2: Directed acyclic graph.

The feedback arrow from returns to the SNoG state implies that simple cross-sectional correlation should not automatically be interpreted as causal identification.

Possible empirical strategies include:

- event studies;
- instrumental variables;
- difference-in-differences;
- local projections;
- structural VARs;
- state-space models;
- synthetic controls;
- high-frequency identification where appropriate.

23 Local Projections

For horizon h ,

$$R_{i,t+h} = \alpha_{i,h} + \beta_h S_t + \boldsymbol{\theta}'_h \mathbf{Z}_t + u_{i,t+h}. \quad (62)$$

The sequence

$$\{\beta_0, \beta_1, \dots, \beta_H\} \quad (63)$$

provides an impulse-response profile.

24 Machine Learning

Machine learning can be used for prediction without automatically conferring causal interpretation.

Let

$$\hat{R}_{t+1} = f_{\theta}(\mathbf{X}_t, \mathbf{P}_t, \mathbf{M}_t, \mathbf{G}_t). \quad (64)$$

Candidate models include:

- random forests;
- gradient boosting;
- recurrent neural networks;
- temporal convolutional networks;
- transformers;
- state-space neural models;
- ensemble methods.

The central comparison remains

$$\text{Benchmark} \quad \text{versus} \quad \text{Benchmark} + \text{SNoG}. \quad (65)$$

25 Algorithmic Research Pipeline

Pseudo-code

INPUT:

Financial returns
Macroeconomic variables
Geopolitical indicators
SNoG structural variables

STEP 1:

Synchronize observations.

STEP 2:

Clean missing values and detect outliers.

STEP 3:

Construct SNoG state variables.

STEP 4:

Estimate benchmark asset-pricing models.

STEP 5:

Estimate SNoG-augmented models.

STEP 6:

Estimate conditional covariance dynamics.

STEP 7:

Perform in-sample diagnostics.

STEP 8:

Conduct rolling and expanding-window forecasts.

STEP 9:

Evaluate statistical and economic performance.

STEP 10:

Perform robustness and placebo tests.

OUTPUT:

Risk-premium estimates

Volatility responses

Forecast performance

Portfolio implications

Network measures

Structural sensitivity analysis

26 Portfolio Choice

Let portfolio weights be $\mathbf{w}_t$.

Expected portfolio return is

$$\mathbb{E}_t[R_{p,t+1}] = \mathbf{w}_t' \mathbb{E}_t[\mathbf{R}_{t+1}]. \quad (66)$$

Conditional variance is

$$\text{Var}_t(R_{p,t+1}) = \mathbf{w}_t' \boldsymbol{\Sigma}_t \mathbf{w}_t. \quad (67)$$

A mean-variance investor solves

$$\max_{\mathbf{w}_t} \left\{ \mathbf{w}_t' \mathbb{E}_t[\mathbf{R}_{t+1}] - \frac{\lambda}{2} \mathbf{w}_t' \boldsymbol{\Sigma}_t \mathbf{w}_t \right\}. \quad (68)$$

If $\boldsymbol{\Sigma}_t$ depends on the SNoG state, then optimal portfolio weights also become state-dependent.

27 Risk Management

A conditional Value-at-Risk measure can be defined by

$$\mathbb{P}_t(R_{p,t+1} < -\text{VaR}_{\alpha,t}) = \alpha. \quad (69)$$

Conditional expected shortfall is

$$\text{ES}_{\alpha,t} = -\mathbb{E}_t[R_{p,t+1} \mid R_{p,t+1} < -\text{VaR}_{\alpha,t}]. \quad (70)$$

An SNoG risk-management layer would test whether

$$\text{VaR}_{\alpha,t}^{\text{SNoG}} \neq \text{VaR}_{\alpha,t}^{\text{Benchmark}} \quad (71)$$

and whether the difference improves realized coverage.

28 Artificial Intelligence and World Models

A learned state representation may be written as

$$\mathbf{z}_t = \phi_\theta(\mathbf{Y}_{1:t}). \quad (72)$$

A learned transition model is

$$\mathbf{z}_{t+1} = f_\theta(\mathbf{z}_t, \mathbf{a}_t, \varepsilon_{t+1}). \quad (73)$$

A decoder produces observable quantities,

$$\hat{\mathbf{Y}}_t = g_\theta(\mathbf{z}_t). \quad (74)$$

The learned representation should be evaluated against explicit econometric state variables rather than assumed to possess economic meaning merely because it predicts well.

29 Multi-Agent Reinforcement Learning

If the framework is extended to strategic agents, let agent i select action $a_{i,t}$ according to policy

$$a_{i,t} \sim \pi_i(a_i \mid s_t). \quad (75)$$

The joint action is

$$\mathbf{a}_t = (a_{1,t}, \dots, a_{N,t}). \quad (76)$$

The global state evolves as

$$s_{t+1} = F(s_t, \mathbf{a}_t, \varepsilon_{t+1}). \quad (77)$$

This framework can be used for computational experiments concerning strategic interaction, but simulated equilibria should not be confused with empirical evidence.

30 Statistical Validation

The model should be subjected to nested and non-nested tests.

For nested models,

$$M_0 \subset M_1. \quad (78)$$

Possible tests include:

- likelihood-ratio tests;
- Wald tests;
- Lagrange-multiplier tests;
- information criteria;
- predictive loss comparisons;
- Diebold–Mariano tests;
- reality-check procedures;
- bootstrap inference.

The evaluation should distinguish statistical significance from economic significance.

31 Robustness Architecture

A credible empirical result should survive variation in:

1. sample period;
2. asset universe;
3. SNoG measurement;
4. benchmark factors;
5. estimation method;
6. frequency;
7. volatility model;
8. treatment of missing observations;
9. crisis definitions.

A generic robustness statistic can be represented as

$$\hat{\beta}^{(j)} = \hat{\beta} + \Delta_j, \quad (79)$$

where Δ_j represents the sensitivity to specification j .

32 Identification of Structural Breaks

Systemic relationships may change over time.

Let

$$\beta_t = \beta^{(0)} + \Delta_\tau \mathbf{1}(t > \tau). \quad (80)$$

Multiple breaks may be represented as

$$\beta_t = \beta_0 + \sum_{j=1}^J \Delta_j \mathbf{1}(t > \tau_j). \quad (81)$$

This is important because geopolitical and financial regimes are not necessarily stationary.

33 A Measurement Hierarchy

The framework distinguishes four levels.

Table 1: Measurement hierarchy

Level	Object	Example
Structural	Latent systemic state	Strategic tension, network configuration
Observable	Measured indicators	Market spreads, volatility, diplomatic events
Financial	Market valuation	Returns, yields, exchange rates
Welfare	Social consequence	Output, consumption, liquidity, welfare loss

34 Information Set and Rational Expectations

Let the information set be

$$\mathcal{F}_t = \sigma(\mathbf{Y}_t, \mathbf{P}_t, \mathbf{G}_t, \mathbf{M}_t). \quad (82)$$

The conditional expectation is

$$\mathbb{E}_t[X_{t+1}] = \mathbb{E}[X_{t+1} \mid \mathcal{F}_t]. \quad (83)$$

The SNoG extension proposes that the information structure can be augmented by a structured state estimate,

$$\mathcal{F}_t^{SNoG} = \sigma(\mathcal{F}_t, \hat{\mathbf{X}}_t). \quad (84)$$

The empirical question is whether

$$\mathbb{E}[L(X_{t+1}, \mathbb{E}[X_{t+1} \mid \mathcal{F}_t^{SNoG}])] < \mathbb{E}[L(X_{t+1}, \mathbb{E}[X_{t+1} \mid \mathcal{F}_t])]. \quad (85)$$

35 Testable Hypotheses

Hypothesis 1. *SNoG state variables contain incremental information about conditional asset returns after controlling for conventional risk factors.*

Hypothesis 2. *SNoG state variables contain incremental information about conditional volatility after controlling for conventional GARCH dynamics.*

Hypothesis 3. *SNoG state variables contain incremental information about cross-asset correlations during systemic episodes.*

Hypothesis 4. *SNoG-augmented models improve out-of-sample forecasts relative to appropriately specified benchmark models.*

Hypothesis 5. *The economic value of SNoG information varies across asset classes, countries, market regimes, and institutional structures.*

36 Potential Falsification Conditions

The framework should be considered empirically unsupported in a particular application if:

1. the SNoG variables are statistically redundant;
2. apparent effects disappear under appropriate controls;
3. coefficients are unstable across reasonable samples;
4. predictive gains disappear out of sample;
5. improvements are attributable solely to overfitting;
6. the variables provide no economic value after transaction costs and implementation constraints.

This falsification principle is central to the scientific use of the framework.

37 Comparison of Model Layers

Table 2: Benchmark and SNoG-augmented architectures

Dimension	Conventional Model	SNoG-Augmented Model
Risk factors	Financial factors	Financial plus systemic factors
Volatility	Historical financial dynamics	Financial dynamics plus systemic state
Geopolitical information	Often external	Explicit state variable
Feedback	Usually limited	Explicitly modelled
Network structure	Usually implicit	Explicitly represented
Asset pricing	Conventional risk premia	Conditional systemic risk premia
Validation	Statistical fit	Statistical, economic, structural, and out-of-sample tests

38 Limitations

Several limitations require explicit recognition.

First, nuclear systemic risk is difficult to observe directly.

Second, measurement of geopolitical variables may contain substantial subjectivity.

Third, financial markets react to information generated by the same political and economic systems being measured.

Fourth, the distinction between prediction and causal inference is fundamental.

Fifth, machine-learning models may improve predictive performance without providing interpretable structural parameters.

Sixth, strategic actors may change behaviour in response to the publication of a risk model.

Seventh, the welfare interpretation of a risk premium is not mechanically determined by the sign of that premium.

39 Research Design

A complete empirical project can follow the architecture below.

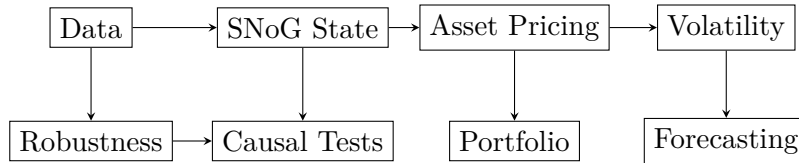


Figure 3: Empirical research design.

40 Expected Contribution

The proposed contribution is a conceptual bridge between two traditionally separated analytical objects:

$$\boxed{\text{Systemic Geopolitical Risk} \longleftrightarrow \text{Financial Risk Pricing}} \quad (86)$$

The key innovation is not the assertion that geopolitical risk affects markets. Such a proposition is too broad to constitute a sufficient scientific contribution.

The stronger contribution is to formulate an explicit state-dependent pricing mechanism:

$$\boxed{\text{SNoG State} \rightarrow \text{Stochastic Discount Factor} \rightarrow \text{Risk Premia} \rightarrow \text{Portfolio Allocation}} \quad (87)$$

combined with the reverse feedback channel,

$$\boxed{\text{Financial Prices} \rightarrow \text{Expectations} \rightarrow \text{Systemic State}}. \quad (88)$$

This produces a coupled financial-geopolitical system.

41 Conclusion

This paper develops a proposed asset-pricing layer for the Standard Nuclear oliGARCHy. The central object is a state-dependent stochastic discount factor whose dynamics depend upon a structured systemic-risk state.

The resulting framework permits analysis of:

1. conditional risk premia;
2. systemic-risk exposures;
3. volatility;
4. covariance and correlation;
5. financial contagion;
6. network centrality;
7. portfolio allocation;
8. sovereign and currency risk;
9. strategic interaction;
10. welfare effects;
11. machine-learning prediction;
12. systemic feedback.

The central empirical test is deliberately demanding:

$$\text{Does SNoG information improve upon established financial models?} \quad (89)$$

If the answer is no, the additional architecture has limited empirical value for the tested application.

If the answer is yes, and the improvement survives appropriate identification, robustness, and out-of-sample tests, the SNoG may provide an empirically useful representation of the interaction between systemic geopolitical risk and financial pricing.

The ultimate objective is therefore not to replace asset-pricing theory, geopolitical analysis, or systemic-risk modelling, but to construct a formally testable interface between them.

References

- [1] K. J. Arrow, *The Role of Securities in the Optimal Allocation of Risk-Bearing*, Review of Economic Studies, 1964.
- [2] F. Black, Studies of Stock Market Volatility Changes, Proceedings of the American Statistical Association, 1976.
- [3] J. Y. Campbell, A. W. Lo, and A. C. MacKinlay, *The Econometrics of Financial Markets*, Princeton University Press, 1997.
- [4] R. F. Engle, Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation, *Econometrica*, 1982.
- [5] R. F. Engle, Dynamic Conditional Correlation, *Journal of Business and Economic Statistics*, 2002.
- [6] E. F. Fama, Risk, Return, and Equilibrium, *Journal of Political Economy*, 1973.
- [7] E. F. Fama and K. R. French, Common Risk Factors in the Returns on Stocks and Bonds, *Journal of Financial Economics*, 1993.
- [8] L. P. Hansen, Large Sample Properties of Generalized Method of Moments Estimators, *Econometrica*, 1982.
- [9] L. P. Hansen and R. Jagannathan, Implications of Security Market Data for Models of Dynamic Economies, *Journal of Political Economy*, 1991.
- [10] C. R. Harvey and Y. Liu, Lucky Factors, *Journal of Financial Economics*, 2016.
- [11] O. Jordà, Estimation and Inference of Impulse Responses by Local Projections, *American Economic Review*, 2005.
- [12] R. E. Lucas, Asset Prices in an Exchange Economy, *Econometrica*, 1978.
- [13] R. C. Merton, An Intertemporal Capital Asset Pricing Model, *Econometrica*, 1973.
- [14] R. J. Shiller, Do Stock Prices Move Too Much to Be Justified by Subsequent Changes in Dividends?, *American Economic Review*, 1981.
- [15] S. A. Ross, The Arbitrage Theory of Capital Asset Pricing, *Journal of Economic Theory*, 1976.
- [16] H. Shefrin, *Beyond Greed and Fear*, Oxford University Press, 2000.
- [17] A. Tversky and D. Kahneman, Judgment under Uncertainty: Heuristics and Biases, *Science*, 1974.
- [18] H. R. Varian, Divergence of Opinion in Complete Markets, *Journal of Finance*, 1985.

Glossary

Asset Pricing

The study of how financial assets are valued and how expected returns are related to risk.

Conditional Risk Premium

The expected excess return conditional on the information available at time t .

Covariance

A measure of the joint variation between two random variables.

Endogeneity

A situation in which an explanatory variable is statistically related to the disturbance term, often because of omitted variables, simultaneity, or reverse causality.

Feedback

A mechanism in which the output of a system influences subsequent system states.

GARCH

Generalized Autoregressive Conditional Heteroskedasticity, a class of models for conditional variance dynamics.

Network

A collection of nodes connected by weighted relationships.

Risk Premium

Compensation associated with bearing a particular source of risk.

SNoG

Standard Nuclear oliGARCHy, the structural systemic-risk architecture considered in this paper.

State Variable

A variable summarizing information relevant for determining the future evolution of a dynamic system.

Stochastic Discount Factor

A random variable that prices contingent future payoffs through conditional expectation.

Systemic Risk

Risk associated with the deterioration or failure of an interconnected system rather than an isolated entity.

Volatility

A statistical measure of the variability of an asset return or other random variable.

The End